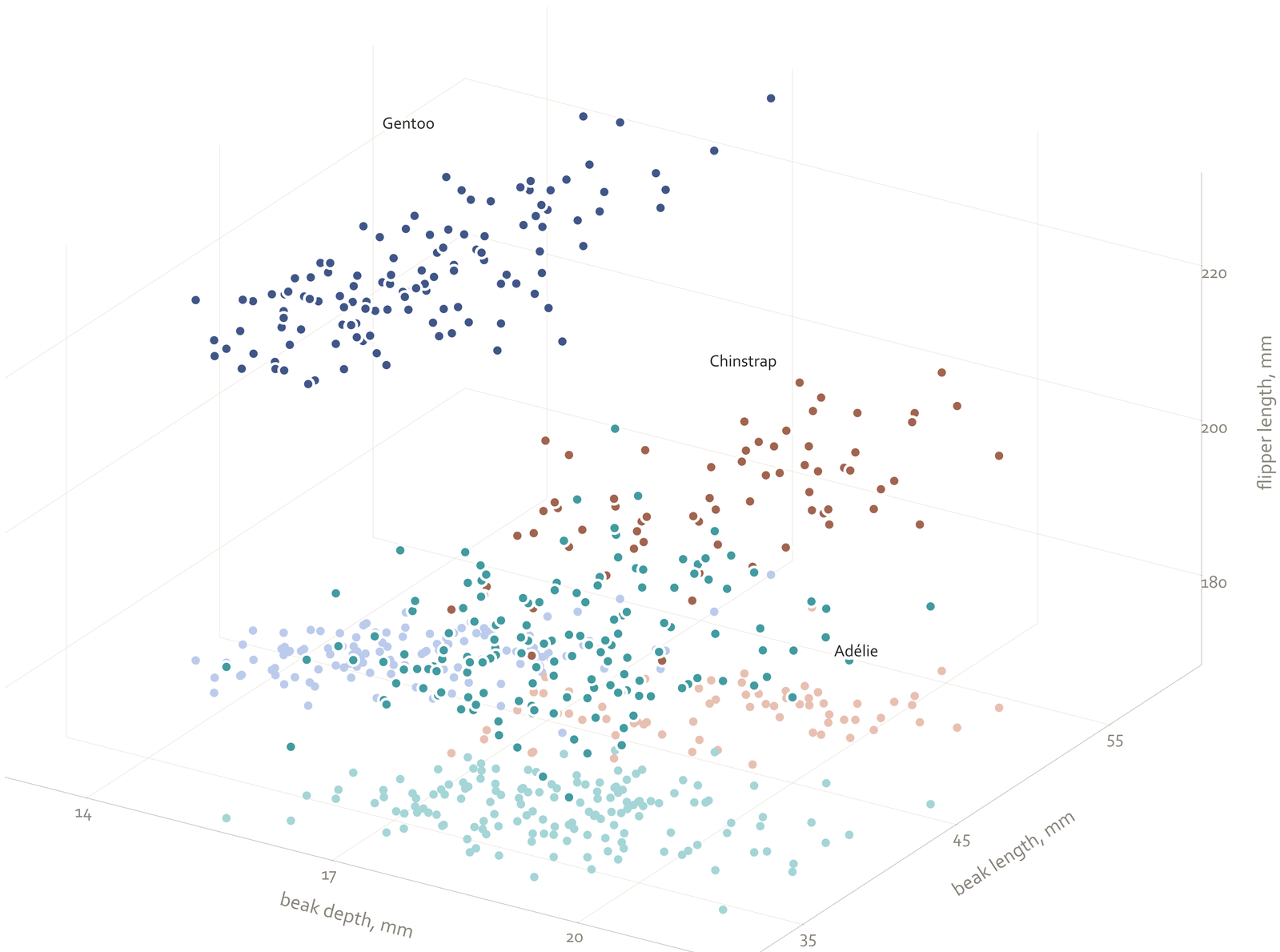
Beak shape, not body size, tells penguin species apart

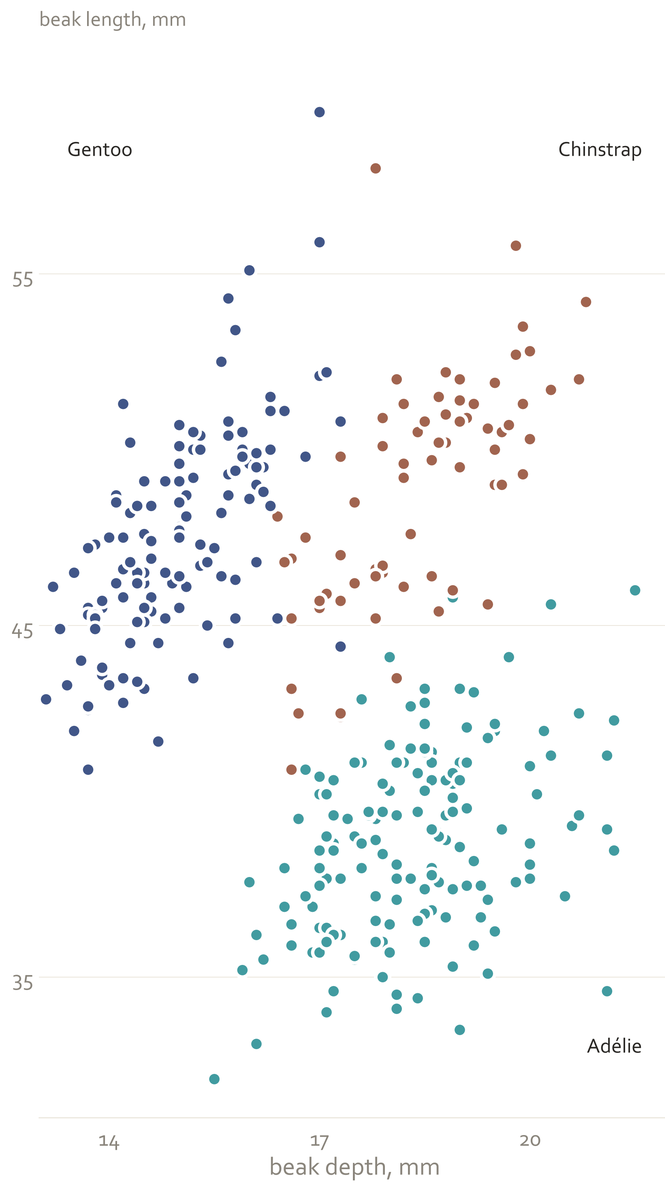
A. Kaminas · Vellum example poster · data: Gorman, Williams and Fraser (2014), Palmer Station LTER

Each dot is one bird. In a, the pale dots on the floor are the same birds seen from above: that view is b.

a Flipper length, a measure of size, sets only Gentoo apart



b Beak length and depth alone separate all three



The question

1. Three penguin species breed on the Palmer Archipelago: Adélie, Chinstrap and Gentoo.
2. Gentoo are heavy, 5 076 g on average; Adélie and Chinstrap differ by only 32 g.
3. Which measure tells all three apart: the size of the body, or the shape of the beak?

What was done

1. 342 birds measured at Palmer Station: beak length and depth, flipper length, body mass.
2. Each measure scaled to mean 0 and standard deviation 1.
3. Each bird left out in turn and assigned to the species with the nearest mean.
4. Once on beak length and depth, once on body mass alone.

What it means

1. Beak length and depth place 94 % of birds in their own species; body mass places 63 %.
2. Adélie beaks are short and deep, Chinstrap long and deep, Gentoo long and shallow.
3. Size sets only Gentoo apart: flipper length alone places 77 %.
4. Beak shape is the measure that tells all three apart.

94 % of birds placed in their own species by beak length and depth, against 63 % by body mass alone

Contact: A. Kaminas · akaminas.github.io/vellum
Data: Gorman, Williams and Fraser (2014), PLoS ONE 9(3): e90081; Palmer Station LTER; via the palmerpenguins R package and vega-datasets 2.8.0; CC0
Accuracies: nearest species mean, leave-one-out, on 342 birds · make_poster_figures.py · poster.tex · git 99753b9-dirty